INAUGURAL CLASS · SECOND SEMESTER 2026

Computational Physics I

Numerical methods and scientific computing for physics.

Yachay Tech University · School of Physical Sciences and Nanotechnology

Physics · ECFN1015 · Level 7 · In person

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The course at a glance

Structure of the course

176
total hours

64
contact hours

6
units

24
topics

Course description

An introduction to the methods and techniques of computational physics, and to their application in current scientific computing.

Prerequisites

Formal requirements and assumed background

Assumed background

- Calculus and linear algebra
- Basic notions of algorithms
- Prior programming experience, in any language
- Physics from the first six semesters
- Experience debugging code

P **Numerical Methods**
ECMC1006

P **Intro to Programming**
ECMC1008

0 **No corequisites**
None required

Why computational physics?

Theory, experiment and computation

I

Theory

Predicts what should happen.

Analytic solutions exist only for a limited set of idealised problems.

II

Experiment

Measures what does happen.

Often expensive, noisy, or inaccessible in the regime of interest.

III

Computation

Solves the equations numerically.

A controlled numerical experiment on a fully specified model.

Learning outcomes

What the course is designed to develop

1. **Scientific programming**
   Object-oriented programming for scientific computing in Linux environments.

2. **Data analysis and visualisation**
   Algorithms and software in Python for analysis, fitting and visualisation.

3. **Modelling of physical systems**
   Numerical methods, image analysis and Fourier analysis.

4. **Research applications**
   Applications in astrophysics, electromagnetism, classical and quantum mechanics.

Course structure

Six curricular units

1 Introduction to computer science

Linux, editors, algorithms

2 Data analysis and visualisation

Python, fitting, approximation

3 Linear algebra and images

Matrices, eigenvalues, arrays

4 Numerical calculus

Derivatives, integrals, optimisation

5 Fourier analysis

DFT, FFT, filters, smoothing

6 Special topics

Wavelets, Monte Carlo, ODEs

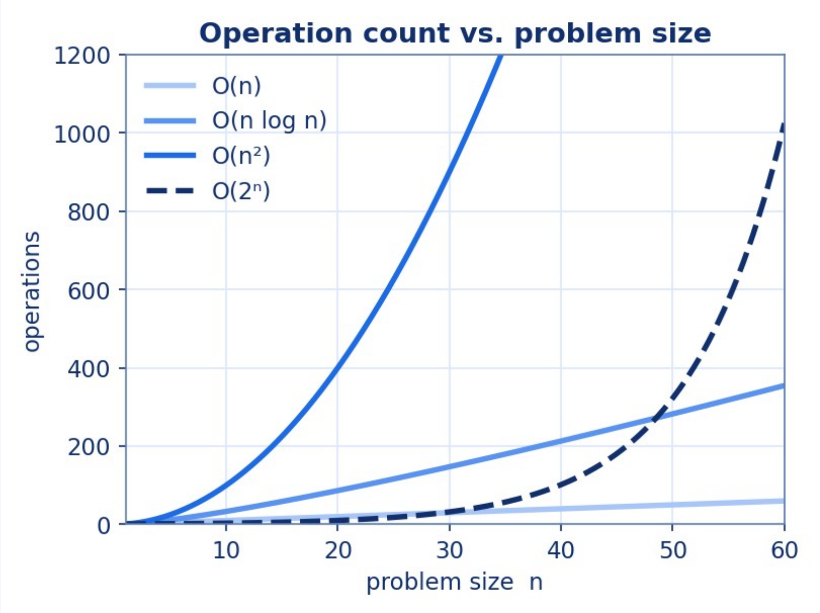
1

UNIT 1

Introduction to computer science

1. Linux operating systems
2. Code editors and environments
3. Scientific programming
4. Algorithms and languages

Set up your working environment during the first week.



Operation count as a function of problem size, for four complexity classes.

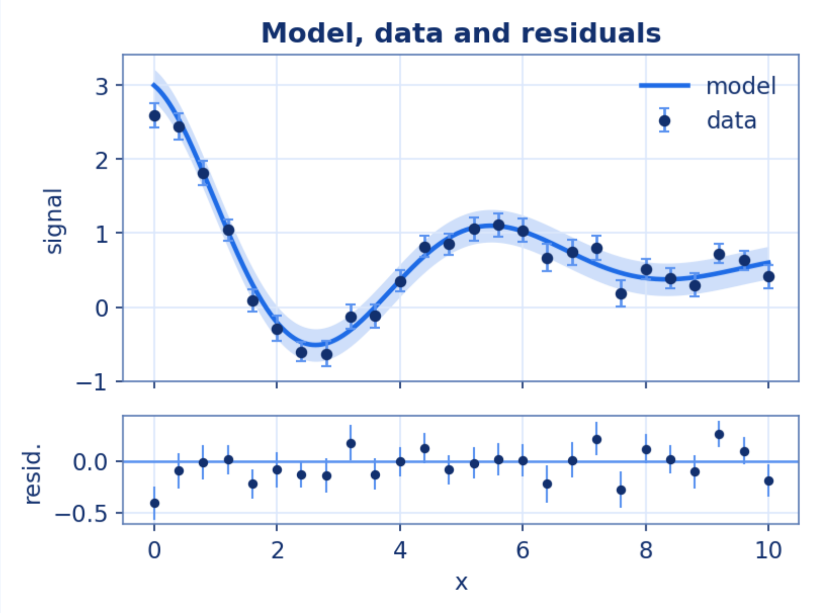
Own figure

2

UNIT 2

Data analysis and visualisation

1. Python basics; errors and uncertainties
2. Procedural vs. object-oriented programming
3. Statistics, fitting and regression
4. Interpolation and splines



Data with uncertainties, a fitted model, and the corresponding residuals.

Own figure

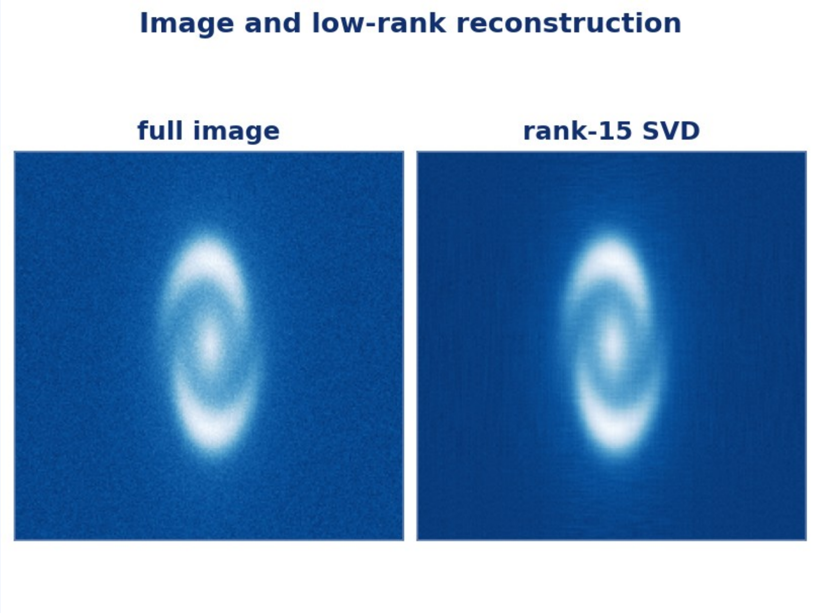
Includes the treatment of uncertainties in reported results.

3

UNIT 3

Linear algebra, matrices and images

1. Linear and non-linear systems
2. Eigenvalue problems
3. Arrays, vectors, matrices, images
4. Data formats and animations



A synthetic image and its rank-15 singular value decomposition.

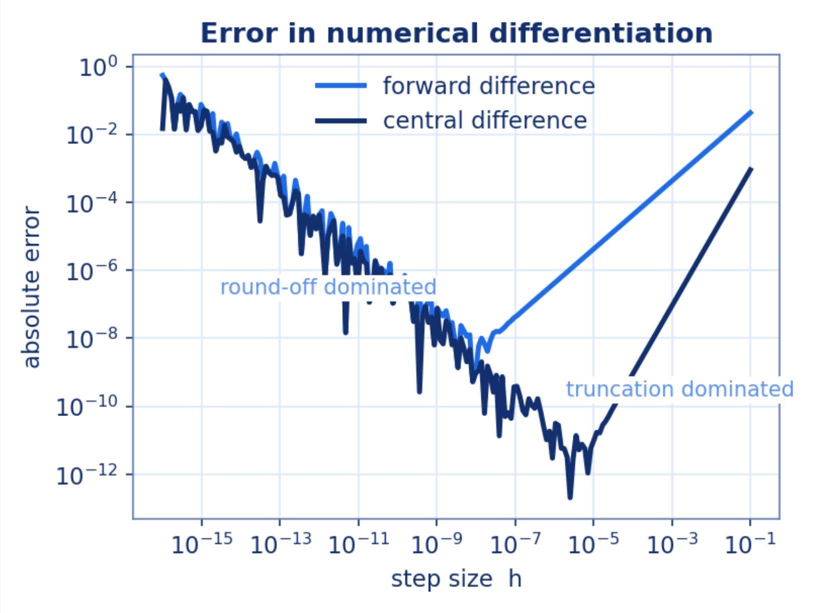
Own figure

Images are treated as matrices throughout this unit.

UNIT 4

4 Numerical calculus and optimisation

1. Numerical differentiation
2. Numerical integration
3. N-dimensional grids and data cubes
4. Optimisation and root finding



Truncation and round-off error in forward and central difference formulae.

Own figure

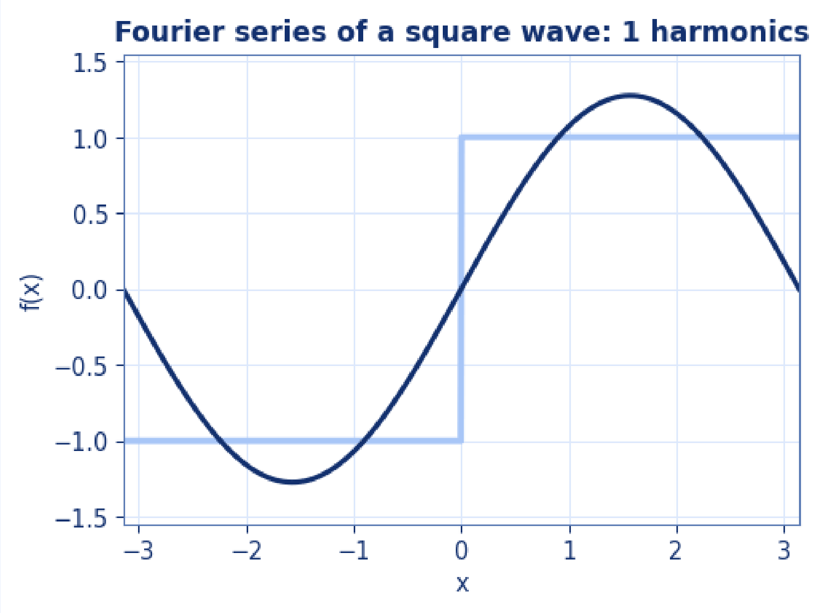
Discretisation and error analysis on finite grids.

5

UNIT 5

Fourier analysis and applications

1. Fourier series
2. DFT and the FFT algorithm
3. Filter design and noise removal
4. Smoothing and edge detection



Partial sums of the Fourier series of a square wave.

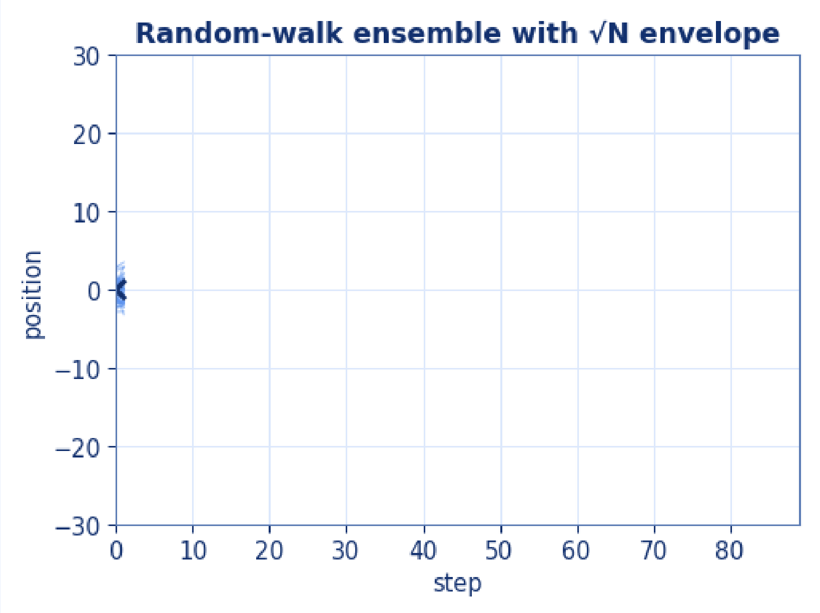
Own figure

Signal analysis in the frequency domain.

UNIT 6

6 Special topics in computational physics

1. Wavelet transforms
2. Monte Carlo methods
3. Random walks and radioactive decay
4. ODEs and initial value problems



An ensemble of random walks and the √N envelope.

Own figure

Topics that connect to research projects in the School.

Teaching methodology

Course delivery and working tools

Course delivery

- Each topic is followed by its implementation
- Every topic ends in working code
- Defences are written and closed-book
- Weekly tutoring hours are available
- Bring a laptop if you can

L

Linux

Working environment

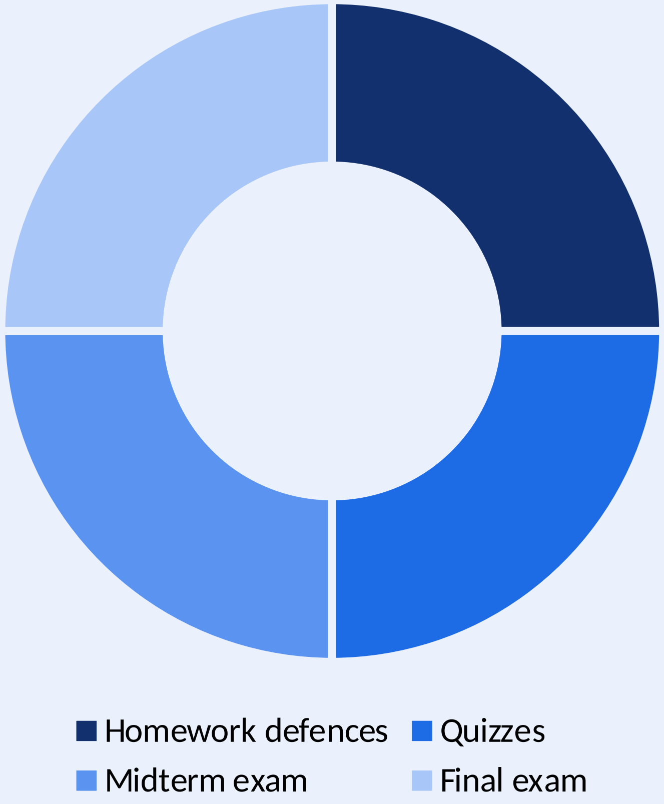
P

Python

Main language

Assessment

Two terms of the period, 50% each



25% **Homework defences**
Report and written defence

25% **Quizzes**
Physics, mathematics and programming

25% **Midterm exam**
First term of the period

25% **Final exam**
Second term of the period

Composition of each component

All defences are written and closed-book

25p Quizzes

- Closed-book, written
- Concepts, physics, mathematics, programming
- At least 4 during the semester

25p Homework defences

- 5p — complete, submitted report
- 20p — closed-book, written defence
- On your report and code · at least 2

25p Midterm exam

- 10p — quiz-like: closed-book, written
- 5p — complete, submitted notebook
- 10p — closed-book, written defence

25p Final exam

- 10p — quiz-like: closed-book, written
- 5p — complete, submitted notebook
- 10p — closed-book, written defence

Calendar

Second semester 2026

Classes	**Every week** Tue 13:00–15:00 · Wed 09:00–11:00 · Fri 13:00–15:00
Tutoring	**Every week** Tue 15:00–16:00 · Wed 11:00–12:00 · 2nd floor, Senescyt Building, or Zoom
30 Sep – 8 Oct	**Midterm exam** First term of the period
7 – 10 Dec	**Final exam** Second term of the period
Announced in class	**Quizzes and defences** Once fixed, they are hard deadlines

Regulations

UITEY Academic Regulations

70

Attendance

70% minimum to pass. Absences justified within 5 days. (Art. 82–83)

6

Make-up exam

A final average of 4.8–5.9 qualifies. Passing raises the grade to 6.0. (Art. 61)

!

Academic integrity

No copying from other students, solution manuals, previous years' solutions, or AI chatbots.

=

Deadlines

Agreed in class; once fixed, they are hard deadlines.

Attendance and late submissions

Course policy

M

Attendance is recorded on Moodle

I record it myself after every class. Please check it regularly and let me know if there is an issue.

70

70% attendance is required to pass

As established by the Vicerrectorado of Yachay Tech and the University regulations.

5

Absences must be justified within 5 days

Official documentation only: the medical centre or Bienestar Universitario, as the regulations establish.

%

Late submissions

With appropriate justification, no penalty. Without it, a penalty of −1% per late hour applies.

Bibliography

Main and complementary sources

Main

Computational Physics: Problem Solving with Python — 2015

Rubin Landau · 3rd ed. · Wiley-VCH

Python Programming and Numerical Methods — 2020

Kong, Siauw & Bayen · 1st ed. · Elsevier · available online

Complementary

An Introduction to Computational Physics — 2006

Tao Pang · 2nd ed. · Cambridge University Press

Recommendations

How to approach the course

1

Submit your OWN work

Write code every week.

Steady weekly work is more effective than concentrated effort before a deadline.

2

Be able to explain what you submit

Defences are closed-book and written. You must be able to account for any code you hand in.

3

Ask questions early

Bring the error message, the input and the expected result to tutoring hours.

Course materials

Scan the code or type the address

Course repository

github.com/wbandabarragan/
computational-physics-1

Previous semesters (archive)

github.com/wbandabarragan/
computational-physics-1-arxiv

Let's get started.

Bring a laptop if you can.

First session: Linux, editors and the working environment.

Lecturer

Wladimir E. Banda-Barragán

Tutoring

Tue 15:00–16:00
Wed 11:00–12:00
2nd floor, Senescyt Building, or Zoom

Classes

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Wed 09:00–11:00
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